Chapter 10: Fitting Regression Models

Regression method is a general method that relates the "**dependent**" variable y to one or more "**independent**" variables $x_1, x_2, \ldots, x_k$.

We often used **multiple regression model** for experimental data through the semester.

Since there is a **strong interplay** between design of experiment and regression analysis, we will discuss regression analysis in this chapter.

The general form of a regression model of independent variable is given by

$$Y = f(X_1, X_2, \ldots, X_k; \beta_0, \beta_1, \ldots, \beta_k) + \varepsilon$$

For some function $f(x_1, x_2, \ldots, x_k)$ and random error $\varepsilon \sim N(0, \sigma^2)$.

The function $f(x_1, x_2, \ldots, x_k)$ is called the **expectation function**.

Most common choice of function f is a linear function of some parameters $\beta_0, \beta_1, \ldots, \beta_k$.

The general form of a **multiple linear regression model** of independent variable is given by

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_k X_k + \varepsilon$$

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$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_k X_k + \varepsilon$$

The term **linear** is used because the above equation is a linear function of the unknown parameters $\beta_0, \beta_1, \ldots, \beta_k$.

The parameter $\boldsymbol{\beta_0}$ is the **intercept** and parameter $\boldsymbol{\beta_j}$ represents the expected **change in the mean response μ_Y per unit change in x_j when all of the remaining independent variables are held constant.**

For this reason the parameters are often called **partial regression coefficients**.

Estimating Model Parameters: The method of **least squares** can be used to estimate the regression coefficients in the regression model.

Suppose $n > k$ observations are available, let y_i denotes the ith observed response and x_{ij} denotes the ith observation of independent variable X_j. The data will appear in the following table:

Observation	Response	Regressors			
i	Y	X_1	X_2	$\cdots$	X_k
1	y_1	x_{11}	x_{12}	$\cdots$	x_{1k}
2	y_2	x_{21}	x_{22}	$\cdots$	x_{2k}
.	.	.	.		.
.	.	.	.		.
.	.	.	.		.
n	y_n	x_{n1}	x_{n2}	$\cdots$	x_{nk}

Consider the following multiple linear regression model:

$$Y_i = \beta_0 + \beta_1 X_{i1} + \beta_2 X_{i2} + \cdots + \beta_k X_{ik} + \varepsilon_i$$

The **least-squares estimates** for the model parameters are obtained by **minimizing** the sum of squares residuals where

$$SS_{\text{Residual}} == \sum_{i=1}^{n} (y_i - \beta_0 - \beta_1 x_{i1} - \beta_2 x_{i2} - \cdots - \beta_k x_{ik})^2$$

The estimates are easily expressible in matrix notation, but that is beyond the scope of this course.

In this class, we rely on R to compute the estimates.

Example 1: Does systolic blood pressure increase with age?

A sample of 27 individuals was randomly selected from a population with ages ranging from 20 to 70 years.

The response variable is measured as the **average systolic blood pressure** at a fixed time during the day over a 1-week period. The explanatory variables are **Age**, **BMI** = Body Mass Index = (weight/(height^2)*703 and **Gender** (1 = Female, 0=Male)

The data are given in the SPSS file on Eclass.

Save the file on your computer. You can use **read.spss** to open the file in R.

You may also use **file.choose()** to find the path to the file. The system will open a window when you enter **file.choose()** for the file access; you can look for your file in the folder where you have earlier archived it. R will return the path to the file.

R codes to Open a SPSS file in R:

```
> library(foreign)

> file.choose()

> data = data.frame(read.spss( "Give the path"))

> head(data)
```

	bldpress	age	BMI	Gender
1	116	34	24	1
2	112	26	32	0
3	153	51	35	0
4	158	58	38	0
5	122	34	30	1
6	125	40	28	1

R Codes to Find the Estimates of Parameters:

```
> attach(data)
> model = lm(bldpress~age+BMI+Gender)
>summary(model)
```

```
Call:
lm(formula = bldpress ~ age + BMI + Gender)

Residuals:
     Min       1Q   Median       3Q      Max 
-10.1749  -2.3855  -0.0917   2.1554   6.7068 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept)  63.2976     5.3923  11.738 3.43e-11 ***
age           0.9986     0.1051   9.503 1.98e-09 ***
BMI           0.9704     0.2125   4.566 0.000137 ***
Gender       -2.0168     1.8519  -1.089 0.287418    
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 4.22 on 23 degrees of freedom
Multiple R-squared:  0.9531,    Adjusted R-squared:  0.9469 
F-statistic: 155.7 on 3 and 23 DF,  p-value: 2.039e-15
```

R Code to Find ANOVA Table:

>anova(model)

```
Analysis of Variance Table

Response: bldpress
          Df Sum Sq Mean Sq F value    Pr(>F)
age        1 7745.3  7745.3 434.933 < 2.2e-16 ***
BMI        1  550.3   550.3  30.902 1.179e-05 ***
Gender     1   21.1    21.1   1.186    0.2874
Residuals 23  409.6    17.8
```

R Codes to Find the Estimates of Parameters:

> model2 = lm(bldpress~Gender)

>summary(model2)

```
Call:
lm(formula = bldpress ~ Gender)

Residuals:
    Min      1Q  Median      3Q     Max 
-34.143  -9.379   1.857  13.621  30.857 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept)  146.143      4.627  31.586   <2e-16 ***
Gender       -13.527      6.668  -2.029   0.0533 .  
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 17.31 on 25 degrees of freedom
Multiple R-squared:  0.1414,    Adjusted R-squared:  0.107 
F-statistic: 4.116 on 1 and 25 DF,  p-value: 0.05327
```

R Codes to Perform Two Sample T-test (With Equal Variance Assumption – Two-sided Alternative):

```
> y_male=bldpress[Gender==0]
> y_female=bldpress[Gender==1]

> t.test(y_male,y_female, var.equal=T)
```

```
        Two Sample t-test

data:  y_male and y_female
t = 2.0287, df = 25, p-value = 0.05327
alternative hypothesis: true difference in means is not equal to 0
95 percent confidence interval:
 -0.2056067 27.2605517
sample estimates:
mean of x mean of y
 146.1429  132.6154
```

Example 2 (Example 1 of Chapter 3): Suppose we would like to compare three brands of gasoline.

Case # 1

Brand 1	Brand 2	Brand 3
15	19	22
19	17	17
14	16	19
16	20	18
$\bar{y}_1 = 16$	$\bar{y}_2 = 18$	$\bar{y}_3 = 19$
$s_1 = 2.16$	$s_2 = 1.83$	$s_3 = 2.16$

H_0: $\mu_1 = \mu_2 = \mu_3$ Reduced model: One Mean Model

H_A: μ_1, μ_2, μ_3 Full model: Three Means Model

R Codes to Perform ANOVA:

```
>B1 = c(15,19,14,16)
>B2 = c(19,17,16,20)
>B3 = c(22,17,19,18)
>y = c(B1,B2,B3)
>f = factor(rep(c("Brand 1","Brand 2","Brand 3"), each=4))
> model = lm(y~f)
> anova(model)
```

```
Analysis of Variance Table

Response: y
          Df Sum Sq Mean Sq F value Pr(>F)
f          2 18.667  9.3333  2.2105 0.1656
Residuals  9 38.000  4.2222
```

If you use **model.matrix(~f)**, you will get

```
   (Intercept) fBrand 2 fBrand 3
1            1         0         0
2            1         0         0
3            1         0         0
4            1         0         0
5            1         1         0
6            1         1         0
7            1         1         0
8            1         1         0
9            1         0         1
10           1         0         1
11           1         0         1
12           1         0         1
attr(,"assign")
[1] 0 1 1
attr(,"contrasts")
attr(,"contrasts")$f
[1] "contr.treatment"
```

In fact, R define dummy variables for Brand 2 and Brand 3.

In general, design of experiment is special case of regression analysis when all independent variables are dummy variables.

What Have We Learned?

The relationship between design of experiment and regression analysis was discussed shortly in this chapter.